The Littlest Migrants:

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Abstract

Previous research on migration typically encounters selection bias or endogenous unobserved characteristics. International adoption circumvents this issue since the placement of the child is independent of unobserved characteristics of the child itself or selection of the 'migrant'. using shift-shares to measure exposure to such a shock, I seek to then investigate the effects downstream market effects of child care costs and education spillovers by combining multiple sources to construct a new dataset. This research therefore creates this dataset for future analysis as well as recasts the discussion and analysis around migration.

Keywords: migration, shift-share, exposure, early childhood, family, childcare, education, adoption

1 Introduction

The political climate in the United States and worldwide has recently been one of closed borders and closed minds. A common and very effective political narrative often takes the form of an us-vs-them framing, and this has recently surfaced with increased funding for Immigration and Customs Enforcement, tightened border controls, and more general public antagonism toward immigration as a policy matter. Thus, it seems increasingly pertinent to examine what exactly makes an immigrant and what kinds of real economic impacts migrants have on the local communities that receive them.

Comparisons of movers and stayers are typically not straightforward, since there is usually heterogeneity in unobserved characteristics such as skill, ability, and drive that both affect and are affected by a stayer's decision to remain or a

mover's decision to migrate. Recognizing that statistical correction alone cannot resolve this problem in non-experimental data (Deaton 2026), researchers have increasingly used identification strategies that rely on quasi-random or natural experiments to study the effects of migration on local product and labor markets. Typically this involves labor market analysis of wages, as basic economic theory would suggest that an increased supply of workers N decreases wages w for all workers, but this simple mechanism does not necessarily hold in practice. Thus, it is important to empirically test such theoretical predictions, since the real world is considerably more complicated than models alone would suggest.

This is not to say that models are without merit, but rather that we as researchers should seek to expand and improve upon them. In the case of migration, a variety of methods have been used to more accurately assess the impact of migrant persons on a host community. Previous literature has used shift-share, or "past settlement," instruments to establish causality. These shift-share instruments have been applied to measure local labor market effects of immigration (Card 2009), to investigate supply shocks from China (Autor, Dorn, and Hanson 2013), and to study how robots and automation affect labor markets (Acemoglu and Restrepo 2017). In the context of this study, I seek to reframe international adoption as a migration shock to local U.S. counties in order to study product market effects, specifically local childcare costs and school-level outcomes, as downstream responses to a small but identifiable population shock.

This research is motivated by the idea that using a specific and unusual type of migrant makes it possible to isolate demand-side effects on local markets from the endogenous location and selection decisions that confound standard migration studies. In particular, because adoptees arrive as infants or young children embedded within American families, and because their county of placement is determined by agency-level matching rather than labor demand or network effects, the variation in their geographic distribution plausibly satisfies the exogeneity requirements of a shift-share design in a way that conventional immigrant flows do not.

Thus, in constructing a novel dataset from a variety of sources to examine the effects of a migration shock in this unique setting, I hope to add to our current understanding of migration and its effects on both the migrants themselves and the local economy.

2 Review of Literature

A Migration and Migrants

Researchers studying migration and its effects are often confronted with selection bias when trying to identify causal relationships. As pointed out by Borjas 1987, the endogeneity of the migration decision creates a selection problem that makes clean causal inference difficult from observational data.

To address this endogeneity, researchers have turned to natural or quasi-random experiments as instruments for migration decisions, for example lottery systems (McKenzie, Stillman, and Gibson 2010). A follow-up study examined the long-term impacts of moving from a poor to a rich country (Gibson et al. 2015). Forced migration and random resettlement have also served as natural experiments (Card 1999). These designs sidestep the selection bias inherent in voluntary migration by making the treatment, i.e. the decision to move, independent of the covariates and error term that would otherwise confound it.

These studies test longstanding beliefs about what opening or closing borders may mean in practice. They test the classic labor market model under a change in labor supply, or they investigate whether there are real returns to moving to a richer country, that is, whether the grass is in fact greener on the other side.

I seek to enrich and broaden this body of research by interrogating what exactly is meant by the classical definition of an immigrant, and by investigating the effect of small, plausibly exogenous migration shocks on specific local product markets, namely childcare and elementary schooling, that are directly downstream of young children's presence in a community.

B International Adoption

Broadly, the academic literature on adoption has approached the subject through psychology (studying belonging and child health, Zeleke, Koester, and Lock 2018; Moroney et al. 2025), sociology (studying openness and the American family, Briggs 2012; Trenka, Oparah, and Shin 2021), and demography, where intercountry adoption has been termed "quiet migration" (Weil 1984). The demographic framing applies specifically to intercountry adoption, in which children from one country are adopted into another, a process that functions somewhat distinctly from domestic adoption and the foster care system. In this vein, Leinaweaver 2014 argues that although adoption may be viewed as a type of migration, it differs in important respects: in volume it is typically

much smaller than labor migration, and it involves only children rather than working-age adults or entire households.

International adoption in the United States peaked in 2004, when the State Department recorded upwards of 20,000 adoptions (Bates and Gramlich 2025), with an overall total of roughly 285,000 children adopted between 1999 and 2024.

The economics literature incorporating adoptees has broadly focused on isolating "place-based" versus "genetic" effects (Sacerdote 2007; Chiovelli et al. n.d.). This framework distinguishes the uprootedness effect, i.e. the general impact of displacement, which can create incentives to invest in portable assets like education, from exposure effects, which capture the impact of the destination's specific level of development. The present study draws on this tradition but redirects attention from adoptee outcomes to the market-level responses of the receiving community itself.

i. A Caveat

In this research, I use international adoption as an instrument to proxy migration in order to work around the selection problem endemic to studying immigration decisions. That said, several distinctions must be drawn between "typical" migration, e.g. labor migration, refugee flows, or other forced displacement, and adoption.

One distinction concerns age and physical health. Adoptees are typically young children or infants integrated into a new family in the host country. This contrasts with the typical labor migrant profile, which tends toward working-age adults migrating alone or with partial family units, often selected on education or skill for visa qualification. A second, related point concerns health selection: there is a common misconception that internationally adopted children are selected for higher-than-average physical health. In fact, disability rates among international adoptees are comparable to those of domestically adopted children and slightly higher than the general domestic population (Kreider and Cohen 2009). This is actually useful for the present study's product-market focus: elevated rates of developmental assessment and early intervention referrals among adoptees strengthen the theoretical case that their arrival generates measurable demand in pediatric and early-childhood services, including the school-based special education and support services that form one of the two outcome categories studied here. Depending on exploratory analysis, one path forward is to

condition on disability status, or to verify whether the disability-rate finding in Kreider and Cohen 2009, whose analysis was conducted using the 2000 Census, holds in more recent cohorts.

3 Methodology

A The Data

i. State Department Adoption Statistics

There will be three main data sources. The first is State Department data on adoption statistics. From 2008 to 2024, the U.S. State Department released annual reports on intercountry adoption statistics,[1] including country of origin, sex of the child, state of placement, and visa type. The start of these reports coincides with U.S. implementation of the Hague Convention on Intercountry Adoption, which created a more regulated and consistently documented intercountry adoption framework and marks the beginning of the most reliable panel data on adoption flows. This implementation also introduces structural breaks in adoption flows that can serve as additional identifying variation.

ii. The National Survey of Adoptive Parents

The National Survey of Adoptive Parents (NSAP)[2] is the first large-scale, nationally representative survey of families with adopted children in the U.S. Conducted by the CDC and DHHS, it collected data from 2,089 adoptive parents to capture family characteristics, adoption experiences, and child well-being across foster care, private domestic, and international adoptions. For the purposes of this study, the NSAP is particularly useful for characterizing the baseline socioeconomic profile of adoptive families, including income, education, and geographic location, which will inform the construction of synthetic comparison counties described in Section D.

iii. U.S. Census Data

The United States has historically maintained strong demographic data infrastructure. The 5-year American Community Survey (ACS) provides race, sex,

[1] `https://travel.state.gov/content/travel/en/Intercountry-Adoption/adopt_ref/AnnualReports.html`

[2] `https://aspe.hhs.gov/national-survey-adoptive-parents-nsap`

age, household income, number of children, and related characteristics at the county level, making it the primary source for constructing the baseline covariate index used in synthetic control matching.

In addition, restricted-access Census data may allow internationally adopted children to be traced directly. Kreider and Cohen 2009 used the 2010 Census long form to trace such children for a study of disability rates. I would therefore aim to construct a comparable sample using the 1990, 2000, 2010, and 2020 long-form data. The 2020 Census had well-documented data collection errors stemming from COVID-19 disruptions to enumeration; rather than excluding 2020 entirely, I plan to evaluate the severity of these errors in the specific counties and demographic cells relevant to this study and to apply differential weighting or partial exclusion on a principled basis rather than a blanket one.

iv. Product Market Outcome Data

For childcare costs, the primary source is the National Database of Childcare Prices (NDCP), published by the U.S. Department of Labor Women's Bureau, which provides county-level annual estimates of childcare prices by age group and care type for 2008–2018.

For school-level outcomes, administrative data from the Common Core of Data (CCD), published annually by the National Center for Education Statistics (NCES), provides school- and district-level enrollment counts, student-to-teacher ratios, per-pupil expenditure, and special education and ELL classification rates at the county level over a panel long enough to span both the Guatemala and China shocks. In particular, I focus on per-pupil instructional spending net of formula-driven adjustments, which avoids the mechanical link between enrollment counts and state funding formulas that would otherwise confound identification.

v. Potential Adoption Agency Records

International adoption in the United States is a heavily regulated process involving social workers, adoption agencies, the receiving country, the sending country, and the parents and child themselves. Consequently, there exists a body of court documents, case files, and immigration paperwork for each adopted child. I plan to contact adoption agencies active during the relevant period to request access to this data. Pending approval, this would meaningfully enrich the analysis by incorporating primary sources; however, the core analysis does

not depend on it, and it should be treated as a potential supplement rather than a requirement.

If this data becomes available, sentiment analysis and natural language processing could be explored, since required adoption paperwork typically includes textual material such as life histories and prospective-parent interviews. This could yield insight into how adoptive parents' expressed motivations or characteristics vary across sending countries or over time, potentially as a robustness check on the NSAP-based family profiles or as a future extension of the current research.

vi. A Novel Dataset

Taken together, I hope to construct a county-year panel linking adoptee inflows (from State Department and Census sources), adoptive family characteristics (from the NSAP and ACS), and product market outcomes (from the NDCP and CCD). This dataset, together with the analysis that follows, will be a central contribution of this research. In constructing it, I aim to better understand the mechanisms behind household decision-making in response to demographic change, and to provide what is, to my knowledge, the first systematic county-level panel linking intercountry adoption flows to downstream childcare and school-market outcomes.

B Discontinuities in Policy Shifts

In this proposal, I use adoption policy shifts in sending countries as exogenous shocks to the receiving country. For example, between 1997 and 2008 just under 30,000 children were adopted from Guatemala, but because Guatemala did not ratify the Hague Convention on Intercountry Adoption in a manner satisfying U.S. implementation requirements, the adoption pipeline effectively ended on December 31, 2007. This closure was driven by Guatemalan domestic politics and adoption-system irregularities, and is plausibly independent of the placement or location of any individual child within the United States.

A second example is the relaxation of China's strict one-child policy in 2015, which corresponded with a substantial drop in adoptions from China, from 2,354 children in the year of the policy change to just 819 four years later. This shift was driven by China's internal demographic and fertility trends, and is plausibly exogenous to any individual child's placement within the United States.

Together, these two shocks provide variation across two distinct sending-country contexts (Latin America and East Asia), two different time periods, and two different shock types, namely an abrupt pipeline closure in the case of Guatemala and a gradual policy-driven decline in the case of China. This diversity of identifying variation strengthens the overall identification argument by reducing reliance on any single event.

C The Model

For our model, we use the base framework given by Bartik 1991, as expanded by Borusyak, Hull, and Jaravel (BHJ) (Borusyak, Hull, and Jaravel 2025), to construct shift-share instruments measuring exposure to adoptee inflows. This methodology exploits exogenous shocks in the changing regulatory landscape of international adoption without requiring assumptions about the exogeneity of adoptee counts in any given local area, a quantity that is likely endogenous given network effects and budget constraints among families seeking to adopt internationally.

We estimate two specifications, one for each outcome category:

$$y_{it}^{(j)} = \beta^{(j)} x_{it} + \gamma^{(j)\prime} \mathbf{w}_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \tag{1}$$

where $j \in \{\text{childcare}, \text{school}\}$ indexes the outcome category, $y_{it}^{(\text{childcare})}$ is the average licensed childcare price per week for ages 0–4 in county i and year t (from the NDCP), and $y_{it}^{(\text{school})}$ is per-pupil instructional spending net of formula-driven adjustments (from the CCD). $\beta^{(j)}$ is the coefficient of interest for each outcome, x_{it} is the treatment variable, i.e. the exposure of county i in year t to adoptee inflows, $\mathbf{w}_{it}$ is a vector of county-level controls, and α_i and δ_t are county and year fixed effects, respectively, included to absorb time-invariant county characteristics and common year-level shocks.

The BHJ framework (Borusyak, Hull, and Jaravel 2025) specifically requires a shock-level natural experiment for shift-share instruments to be valid, a requirement satisfied here by the sending-country policy shifts described in Section B. We therefore construct the instrument

$$z_{it} = \sum_{k=1}^{K} s_{ik}\, g_{kt}, \tag{2}$$

where $\{g_{kt}\}$ are the adoption policy shifts for sending country k in year t (such

as China's 2015 policy relaxation or the 2007 Guatemala pipeline closure) and $\{s_{ik}\}$ are the pre-period shares of county i's historically-adopted-child population attributable to sending country k. These shares will be calculated at the county level, since census-tract-level analysis would likely lack sufficient statistical power given the rarity of adoption events, and state-level aggregation would sacrifice too much local variation.

Ideally, the comparison shares would be measured directly as the number of internationally adopted children within each county prior to the study period. However, given the small sample sizes involved, this is likely to raise data-availability and privacy-suppression issues at the county level. As a fallback, I propose using the pre-period share of foreign-born residents under age 5 in each county as a proxy, since most international adoptees are between two and three years of age at placement (or, in the case of Chinese adoptees, more often infants) (Guest 2019). This proxy is imprecise, as the typical foreign-born child under 5 is not the same as an adoptee, and will attenuate estimates toward zero relative to the true effect. The attenuation should therefore be treated as a lower bound on the true coefficient rather than a point estimate, and I plan to validate the proxy against any agency-record data that becomes available.

D Exposure and Exogeneity

The framework we use relies on two assumptions: (i) that shifts are exogenous, and (ii) that shares are exogenous.

The exogeneity of the *shifts* is a relatively safe assumption. Unlike labor migration, the placement of an adopted child in a given county is determined by household-level adoption-agency matching and bureaucratic requirements, not by local labor demand, which weakens the classic shift-share confound in which the shock is correlated with local economic trends. The sending-country policy events used as shocks, namely China's one-child-policy relaxation and Guatemala's non-ratification of the Hague Convention, were driven by entirely domestic political and demographic considerations in the sending country, with no plausible direct link to conditions in specific U.S. counties.

Verifying that the *shares* are exogenous requires more care. Counties that adopt internationally are systematically different from those that do not, being typically wealthier, more internationally connected, and different in baseline demographic composition. To address this, I propose constructing a synthetic comparison group: for each treated county (one with meaningful historical adop-

tion exposure), I will construct a synthetic counterfactual as a weighted linear combination of non-treated counties with similar baseline characteristics, including racial composition, citizen-versus-non-citizen population shares, and median household income. To identify the relevant matching characteristics, I will use LASSO regression or principal component analysis. The standard validation check for parallel pre-trends between treated and synthetic control counties will be reported as a central robustness exhibit for each specification.

The remaining threat to identification is that the policy shocks driving the instrument may be correlated with the outcome variable through channels other than adoptee exposure itself. This threat takes different forms for each outcome.

For childcare costs, the concern is that sending-country policy shifts coincide with U.S. macroeconomic or regulatory changes that independently move childcare prices, for instance minimum wage increases (which directly affect childcare worker compensation), zoning or licensing law changes, or broader demographic fertility trends. The key placebo test is whether counties with no historical adoption exposure from the affected sending country also show childcare price movements around the shock dates. If such counties are unaffected, the shock's effect on prices is plausibly operating through the adoptee channel rather than a national aggregate trend.

For school-level outcomes, the concern is the mechanical link between enrollment and school funding formulas: adding students to a district directly changes formula-driven spending inputs, creating a definitional rather than causal relationship between adoptee presence and spending. This is why the preferred school outcome is per-pupil instructional spending net of formula-driven adjustments rather than gross enrollment-driven budget figures.

4 Implications

In using adoptees as migrant-level shocks, we are able to simultaneously interrogate the definition of a migrant and sidestep some of the identification issues that currently exist in the literature. Measuring product market effects via childcare costs and educational spillovers will allow us to isolate the demand shocks generated by a small, exogenously-placed migrant population from the confounding selection and network effects that plague standard immigration studies.

In studying what happens to local markets when counties receive an exoge-

nous increase in young children through international adoption, I seek to offer a cleaner test of the downstream market effects of migration than has previously been possible, while also challenging prevailing assumptions about who counts as a migrant and what effects migrants of different types can be expected to produce.

AI Disclosure

Artificial Intelligence (specifically Claude) was used to help me find articles in Chinese or from Chinese media. These sources (because my Chinese skills are middling at best) were then pasted into Google Translate into English. Therefore there may be some errors in translation.

I also used Claude to edit my writing, i.e., as a copy editor. Transcripts of the AI chat can be provided upon request.

References

Acemoglu, Daron and Pascual Restrepo (Mar. 2017). *Robots and Jobs: Evidence from US Labor Markets*. DOI: 10.3386/w23285. URL: https://www.nber.org/papers/w23285 (visited on 06/16/2026).

Autor, David H, David Dorn, and Gordon H Hanson (Oct. 1, 2013). "The China Syndrome: Local Labor Market Effects of Import Competition in the United States". In: *American Economic Review* 103.6, pp. 2121–2168. ISSN: 0002-8282. DOI: 10.1257/aer.103.6.2121. URL: https://pubs.aeaweb.org/doi/10.1257/aer.103.6.2121 (visited on 06/14/2026).

Bartik, Timothy J. (Sept. 1, 1991). *Who Benefits from State and Local Economic Development Policies?* W.E. Upjohn Institute. ISBN: 978-0-585-22394-0. DOI: 10.17848/9780585223940. URL: https://research.upjohn.org/up_press/77/ (visited on 06/14/2026).

Bates, Hannah and John Gramlich (July 25, 2025). *International adoptions to the U.S. have slowed to a trickle, matching trends in other countries*. Pew Research Center. Section: Parenthood. URL: https://www.pewresearch.org/short-reads/2025/07/25/international-adoptions-to-the-us-have-slowed-to-a-trickle-matching-trends-in-other-countries/ (visited on 06/13/2026).

Borjas, George J. (May 1, 1987). *Self-Selection and the Earnings of Immigrants*. Rochester, NY. URL: https://papers.ssrn.com/abstract=975941 (visited on 06/13/2026).

Borusyak, Kirill, Peter Hull, and Xavier Jaravel (Jan. 1, 2025). "Design-based identification with formula instruments: a review". In: *The Econometrics Journal* 28.1, pp. 83–108. ISSN: 1368-4221. DOI: 10.1093/ectj/utae003. URL: https://doi.org/10.1093/ectj/utae003 (visited on 06/14/2026).

Briggs, Laura (Mar. 7, 2012). *Somebody's Children: The Politics of Transracial and Transnational Adoption*. Google-Books-ID: Bnxcsv9sWHUC. Duke University Press. 376 pp. ISBN: 978-0-8223-5161-0.

Card, David (1999). "The Causal Effect of Education on Earnings". In: *Handbook of Labor Economics*. Vol. 3. Elsevier, pp. 1801–1863. ISBN: 978-0-444-50187-5. DOI: 10.1016/S1573-4463(99)03011-4. URL: https://linkinghub.elsevier.com/retrieve/pii/S1573446399030114 (visited on 04/27/2026).

— (Apr. 1, 2009). "Immigration and Inequality". In: *American Economic Review* 99.2, pp. 1–21. ISSN: 0002-8282. DOI: 10.1257/aer.99.2.1. URL: https://pubs.aeaweb.org/doi/10.1257/aer.99.2.1 (visited on 06/16/2026).

Chiovelli, Giorgio et al. (n.d.). "Forced displacement and human capital: Evidence from separated siblings". In: ().

Deaton, Angus S (2026). *The Analysis of Household Surveys : A Microeconometric Approach to Development Policy*. World Bank. URL: https://documents.worldbank.org/en/publication/documents-reports/documentdetail/593871468777303124 (visited on 06/18/2026).

Gibson, John et al. (Nov. 16, 2015). "The long-term impacts of international migration : evidence from a lottery". In: *Policy Research Working Paper Series*. Number: 7495. URL: https://ideas.repec.org//p/wbk/wbrwps/7495.html (visited on 06/16/2026).

Guest (June 13, 2019). *Five Decisions Every Adoptive Family Needs to Make*. Holt International. URL: https://www.holtinternational.org/five-decisions-every-adoptive-family-needs-to-make/ (visited on 06/16/2026).

Kreider, Rose M. and Philip N. Cohen (Nov. 2009). "Disability among internationally adopted children in the United States". In: *Pediatrics* 124.5, pp. 1311–1318. ISSN: 1098-4275. DOI: 10.1542/peds.2008-3206.

Leinaweaver, Jessaca B. (2014). "The Quiet Migration Redux: International Adoption, Race, and Difference". In: *Human organization* 73.1, pp. 62–71. ISSN: 0018-7259. DOI: 10.17730/humo.73.1.f1x606xr745wnw4k. URL:

https://pmc.ncbi.nlm.nih.gov/articles/PMC4295828/ (visited on 06/15/2026).

McKenzie, David, Steven Stillman, and John Gibson (2010). "How Important Is Selection? Experimental Vs. Non-Experimental Measures of the Income Gains from Migration". In: *Journal of the European Economic Association* 8.4, pp. 913–945. ISSN: 1542-4766. URL: https://www.jstor.org/stable/25700908 (visited on 01/15/2026).

Moroney, Dearbhla et al. (2025). "Adoption and social identity loss: Insights from adults adopted through Ireland's mother and baby homes". In: *British Journal of Social Psychology* 64.2. _eprint: https://bpspsychub.onlinelibrary.wiley.com/doi/pdf/10.1111/bjs e12869. ISSN: 2044-8309. DOI: 10.1111/bjso.12869. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/bjso.12869 (visited on 06/18/2026).

Sacerdote, Bruce (Feb. 1, 2007). "How Large are the Effects from Changes in Family Environment? A Study of Korean American Adoptees". In: *The Quarterly Journal of Economics* 122.1, pp. 119–157. ISSN: 0033-5533. DOI: 10.1162/qjec.122.1.119. URL: https://doi.org/10.1162/qjec.122.1.119 (visited on 06/13/2026).

Trenka, Jane Jeong, Julia Chinyere Oparah, and Sun Yung Shin (Mar. 23, 2021). *Outsiders Within: Writing on Transracial Adoption*. Google-Books-ID: uX4jEAAAQBAJ. U of Minnesota Press. 524 pp. ISBN: 978-1-4529-6520-8.

Weil, Richard H. (June 1, 1984). "International Adoptions: The Quiet Migration". In: *International Migration Review* 18.2, pp. 276–293. ISSN: 0197-9183. DOI: 10.1177/019791838401800205. URL: https://doi.org/10.1177/019791838401800205 (visited on 06/16/2026).

Zeleke, Waganesh A., Lynne S. Koester, and Gabriella Lock (May 1, 2018). "Parents' Understanding of Adopted Children' s Ways of Being, Belonging, and Becoming". In: *Journal of Child and Family Studies* 27.5, pp. 1428–1439. ISSN: 1573-2843. DOI: 10.1007/s10826-017-0995-y. URL: https://doi.org/10.1007/s10826-017-0995-y (visited on 06/18/2026).